

Course Name:-B.Tech

SEMESTER EXAM, JULY-2022

Semester:- Second

Paper Name:- Fundamental of Electronics Engineering Paper Code:-TES205

समय- 2 घण्टे, 20 मिनट प्रति घंटे अतिरिक्त-दृष्टिवाचित एवं सह लेखक परीक्षार्थियों के लिए।

अधिकतम अंक-70

Instructions:

- The question paper consists of three sections namely A, B, C. All sections are compulsory.
- Section A- Each question carries 04 marks. All questions are compulsory.
- Section B- Answer any 2 out of 4 given questions in maximum one hundred fifty(150) words. Each question carries 10 marks.
- Section C- Answer any 2 out of 3 given questions in maximum two hundred fifty (250) words. Each question carries 15 marks.

निर्देश:

- प्रश्न पत्र में तीन खण्ड अ, ब, व स हैं। सभी खण्ड अनिवार्य हैं।
- खण्ड-अ में प्रत्येक प्रश्न चार अंक का है। सभी प्रश्न अनिवार्य हैं।
- खण्ड-ब में चार प्रश्नों में से किसीदो प्रश्नों के उत्तर अधिकतम डेढ़ सौ (150) शब्दों में दें। प्रत्येक प्रश्न 10 अंक का है।
- खण्ड-स में तीन प्रश्नों में से किसीदो प्रश्नों के उत्तर अधिकतम डेढ़ सौ (250) शब्दों में दें। प्रत्येक प्रश्न 15 अंक का है।

Section - A (खण्ड-अ)

Objective Questions (वस्तुनिष्ठ प्रश्न)

1. Answer all the following questions.

5×4=20

- निम्नलिखित सभी प्रश्न अनिवार्य हैं।
- What is the Gray code word for the binary number 101011?
(a) 101011 (b) 110101 (c) 011111 (d) 111110
 - Addition of pentavalent impurity to a semiconductor creates many.....?
(a) Free Electrons
(b) Holes
(c) Valence electrons
(d) Bound Electrons
 - Total emitter current of transistor is ...?
(a) $I_E - I_C$ (b) $I_C + I_E$ (c) $I_B + I_C$ (d) $I_B - I_C$
 - FET operations depends on?
(a) Minority charge carriers
(b) Majority charge carriers
(c) Majority and Minority charge carriers
(d) None
 - Trivalent impurities are...?
(a) Aluminium(Al), Arsenic(As)
(b) Boron(B), Antimony(Sb)
(c) Indium(In), Aluminium(Al)
(d) None

Section – B (खण्ड-ब)

Short Answer Questions (लघुउत्तरीय प्रश्न)

2. Answer any two of the following questions in maximum 150 words.

2×10=20

निम्नलिखित में से किन्हीं दो प्रश्नों के उत्तर अधिकतम 150 शब्दों में दें।

- i) What do you mean by P-N junction? Explain the phenomenon of the formation of depletion layer in P-N junction. What will be the effect on the depletion layer of a P-N junction diode under different biased condition?
- ii) For a semiconductor diode, define static and dynamic Resistance. Also differentiate between transition and diffusion capacitance of a P-N junction diode.
- iii) Find the voltage at which the reverse current in a Ge diode at room temperature will reach 90% of its saturation value. (a) Find the ratio of diode current with a forward bias of 0.05V to the current with a reverse bias of 0.05V? (b) The reverse saturation current of a Ge diode is $10\mu\text{A}$. Find current under forward biases of 0.1V, 0.2V, 0.3V?
- iv) Differentiate between intrinsic and Extrinsic semiconductors. Also define the terms conductivity and energy band gap of semiconductors.

Section – C (खण्ड-स)

Descriptive Questions (विवरणात्मक प्रश्न)

3. Answer any two of the following question in maximum 250 words.

2×15=30

निम्नलिखित में से किन्हीं दो प्रश्नों के उत्तर अधिकतम 250 शब्दों में दें।

- i) Explain the basic construction of n-channel Enhancement MOSFET. Draw and explain its characteristics. Also differentiate between MOSFET and BJT.
- ii) What is transistor biasing?. List different types of transistor biasing. What do you mean by Fixed or Base transistor biasing.
- iii) What are the main application areas of semiconductor diode. Define the term rectification and draw the circuit and output waveform of Half Wave Rectifier (HWR). Also write all parameters of HWR.

SEMESTER END EXAMINATION, JULY-2022

Course Name: B.Tech.

Semester : IInd

Paper Name: Engineering Mathematics-II

Paper Code: TBS-202

Time: 2 Hrs. + 20 Minutes per hour extra for V.I.
and examinees with writer

Max. Marks: 70

समय - 2 घण्टे + 20 मिनट प्रति घण्टे अतिरिक्त दृष्टिबाधित एवं अधिकतम अंक: 70
सहलेखक परीक्षार्थियों के लिए।

Instructions:

- (i) The question paper consists of three Sections namely A, B and C. All Sections are compulsory.
- (ii) Section-A - Each question carries 4 marks. All questions are compulsory.
- (iii) Section-B - Answer any 2 out of 4 given questions in maximum one hundred fifty (150) words. Each question carries 10 marks.
- (iv) Section-C - Answer any 2 out of 3 given questions in maximum two hundred fifty (250) words. Each question carries 15 marks.

निर्देश:

- (i) प्रश्न-पत्र में तीन खण्ड अ, ब व स हैं। सभी खण्ड अनिवार्य हैं।
- (ii) खण्ड-अ में प्रत्येक प्रश्न 04 अंकों का है। सभी प्रश्न अनिवार्य हैं।
- (iii) खण्ड-ब में 04 प्रश्नों में से किन्हीं 02 प्रश्नों के उत्तर अधिकतम डेढ़ सौ (150) शब्दों में दीजिये। प्रत्येक प्रश्न 10 अंकों का है।
- (iv) खण्ड-स में 03 प्रश्नों में से किन्हीं 02 प्रश्नों के उत्तर अधिकतम डेढ़ सौ (250) शब्दों में दीजिये। प्रत्येक प्रश्न 15 अंकों का है।

Section-A (खण्ड-अ)

Objective Questions (वस्तुनिष्ठ प्रश्न)

(5×4=20)

1. Answer all the following questions:
निम्नलिखित सभी प्रश्न अनिवार्य हैं:

(i) The value of $L\left[t^{-\frac{1}{2}}\right]$ is:

(A) $\sqrt{2}$

(B) $\sqrt{\frac{\pi}{s}}$

(C) $\sqrt{s}$

(D) πs

(ii) If $F[f(x)] = f(s)$ then the value of $F[f(ax)]$ is:

(A) $\frac{1}{a} f(s)$

(B) $\frac{1}{a} f\left(\frac{s}{a}\right)$

(C) $f\left(\frac{s}{a}\right)$

(D) $f(as)$

(1)

TBS-202

(iii) The Eigen values of matrix $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ is:

- (A) 0, 1
- (B) 0, 2
- (C) 0, 3
- (D) 0, 4

(iv) The type of partial equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} + 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = 0$ is:

- (A) elliptical
- (B) parabolic
- (C) hyperbolic
- (D) None

(v) The value of $L^{-1} \left[\frac{1}{s-3} \right]$ is:

- (A) e^{-3t}
- (B) e^{3t}
- (C) e^t
- (D) e^{2t}

Section-B (खण्ड-ब)

Short Answer Questions (लघुउत्तरीय प्रश्न)

2.

Answer any two of the following questions in maximum 150 words:

निम्नलिखित में से किन्हीं दो प्रश्नों के उत्तर अधिकतम 150 शब्दों में दीजिये:

- (i) Verify that the following transformation is a linear transformation:

$$T(a, b) = (a+b, a-b, b) \quad \forall a, b \in R$$

Find the range, rank, null space and nullity.

- (ii) Find the Laplace inverse of:

$$\frac{1}{s^4} - \frac{3s}{s^2 + 16} + \frac{s}{s^2 + 4}$$

- (iii) Find the Fourier transform of:

$$f(x) = \begin{cases} 1-x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$$

$\frac{1}{3}$

2, 2, 3, 2

(iv) Solve $\frac{\partial^3 z}{\partial x^3} - 4 \frac{\partial^3 z}{\partial x^2 \partial y} + 4 \frac{\partial^3 z}{\partial x \partial y^2} = 2 \sin(3x + 2y)$.

Section-C (बहु-स)

Descriptive Questions (विवरणात्मक प्रश्न)

3. Answer any two of the following questions in maximum 250 words; निम्नलिखित में से किसी दो प्रश्नों के उत्तर अधिकतम 250 शब्दों में दीजिये:
- (i) Using Laplace transform, find the solution of the initial value problem $y''(t) + y(t) = t$, Given $y'(0) = 1, y(0) = 0$.
 - (ii) Find the characteristic equation of the matrix:

$$A = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$$

Hence find A^{-1} .

- (iii) State and prove convolution theorem and find $L^{-1} \frac{s}{(s^2 + 25)(s^2 + 16)}$.

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Course Name:- B.Tech (CSE/ECE/EE)

SEMESTER EXAM, JULY-2022

Paper Name:- Engineering Mechanics

Semester:- II

Time - 2 Hrs + 20 minutes per hour extra time for V.I. & examinees with writer.

Paper Code:- TES-203

समय- 2 घण्टे + 20 मिनट प्रति घंटे ओरिजिनल-दृष्टिकोणित एवं साह लेखक परीक्षाधिकारियों के लिए।

Max Marks-70

अधिकतम अंक-70

Instructions:

- The question paper consists of three sections namely A, B, C. All sections are compulsory.
- Section A- Each question carries 04 marks. All questions are compulsory.
- Section B- Answer any 2 out of 4 given questions in maximum one hundred fifty (150) words. Each question carries 10 marks.
- Section C- Answer any 2 out of 3 given questions in maximum two hundred fifty (250) words. Each question carries 15 marks.

निर्देश:

- प्रश्न पत्र में तीन खण्ड अ, ब, व स हैं। सभी खण्ड अनिवार्य हैं।
- खण्ड-अ में प्रत्येक प्रश्न चार अंक का है। सभी प्रश्न अनिवार्य हैं।
- खण्ड-ब में चार प्रश्नों में से किसी दो प्रश्नों के उत्तर अधिकतम डेढ़ सौ (150) शब्दों में दें। प्रत्येक प्रश्न 10 अंक का है।
- खण्ड-स में तीन प्रश्नों में से किसी दो प्रश्नों के उत्तर अधिकतम ढाई सौ (250) शब्दों में दें। प्रत्येक प्रश्न 15 अंक का है।

Section - A (खण्ड-अ)

Objective Questions(वस्तुनिष्ठ प्रश्न)

1. Answer all the following questions.

5×4=20

निम्नलिखित सभी प्रश्न अनिवार्य हैं।

- Point of Contraflexure is known as the point...
 - Where shear force changes its sign.
 - Where bending moment changes its sign.
 - Where shear force is maximum.
 - Where bending moment is maximum.
- C.G. of a solid cone lies on the axis at a height of the total height from the base.
 - One-third
 - half
 - one-fourth
 - none of these.
- $S_{nth} = u + a/2(\dots\dots)$.
 - $2n-1$
 - $n-1$
 - $2n$
 - $n/2-1$
- The forces P and Q act at right angles to each other, then the value of R will be-
 - $R = \sqrt{P^2 - Q^2}$
 - $R = P + Q$
 - $R = P - Q$
 - $R = \sqrt{P^2 + Q^2}$
- If work output denoted as W_o and work input is W_i then mechanical efficiency is-
 - W_i/W_o
 - W_o/W_i
 - $1/W_i$
 - $W_o * W_i$

Section - B (खण्ड-ब)

Short Answer Questions (लघुउत्तरीय प्रश्न)

2. Answer any two of the following questions in maximum 150 words.

2×10=20

निम्नलिखित में से किसी दो प्रश्नों के उत्तर अधिकतम 150 शब्दों में दें।

11. State and prove with diagram $\frac{P}{\sin a} = \frac{Q}{\sin b} = \frac{R}{\sin c}$.

Two forces of magnitudes $3P$, $2P$ respectively acting at a point have a resultant R . If the first force is doubled, the magnitude of the resultant is doubled. Find the angle between the forces.

- ii) Determine the forces in all members of truss shown in Fig. 1.

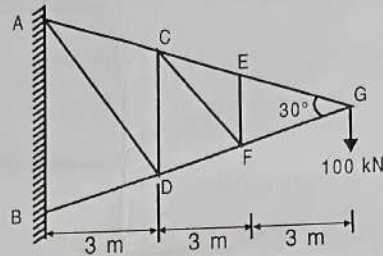


Fig. 1

- iii) Draw the Shear force diagram and bending moment diagram for given beam in Fig. 2. Determine the point of contraflexure also.

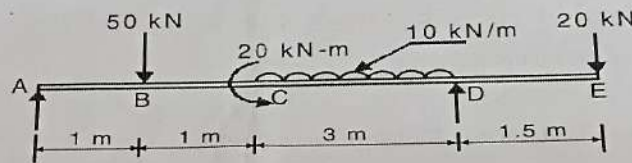


Fig. 2

- iv) A ladder of length L rests against a wall, the angle of inclination being 45° . If the coefficient of friction between the ladder and the ground and that between ground and the wall is 0.5 each, what will be the maximum distance on ladder to which a man whose weight is 1.5 times the weight of ladder may ascend before the ladder begins to slip?

Section - C (खण्ड-स)

Descriptive Questions (विवरणात्मक प्रश्न)

3. Answer any two of the following question in maximum 250 words.

2×15=30

निम्नलिखित में से किन्हीं दो प्रश्नों के उत्तर अधिकतम 250 शब्दों में दें।

What is impulse and impulsive force? State and prove Law of conservation of momentum.

A body has an initial velocity of 10m/sec and is being accelerated at a constant rate of 5m/sec^2 . What will be the velocity after 2 seconds? What distance does it travel during this time?

Define Beam, support and their types.

Solve for the reactions at L, M, N and S on the two cylinders shown in Fig. 3, assuming all surfaces to be smooth.

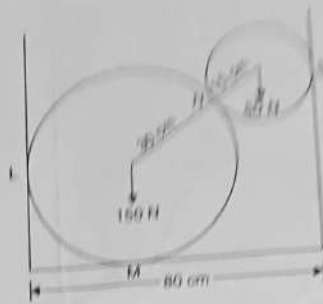


Fig. 3

- iii) Find out the centre of gravity of a solid hemisphere. And Find I_{xx} of the channel section shown in Fig. 4.

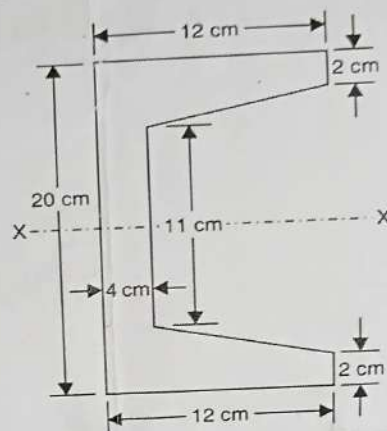


Fig.4

$$I_{xx} = I_{xx1} + I_{xx2} + I_{xx3}$$

joint have a resultant R. If the first
d. Find the angle between the

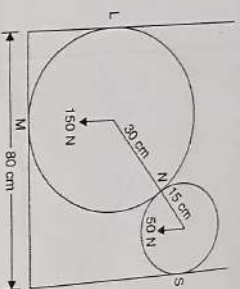


Fig. 3

iii) Find out the centre of gravity of a solid hemisphere. And Find I_{xx} of the channel section shown in Fig. 4.

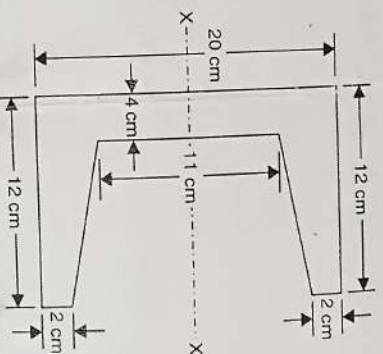


Fig. 4

Handwritten scribbles and marks.

Handwritten: $A \pi R^2$

SEMESTER END EXAMINATION, JULY 2022

Course Name:-B.Tech

Paper Name:-PROFESSIONAL COMMUNICATION

Semester:-IIND

Paper Code:-THS-201

Time - 2Hrs + 20 minutes per hour extra time for V.I. & examinees with writer.

Max Marks-70

समय- 2 घण्टे+ 20 मिनट प्रति घंटे अतिरिक्त-दृष्टिबाधित एवं सहलेखक परीक्षार्थियों के लिए।

अधिकतम अंक-70

Instructions:

- The question paper consists of three sections namely A, B, C. All sections are compulsory.
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- Section B- Answer any 2 out of 4 given questions in maximum one hundred fifty(150) words. Each question carries 10 marks.
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Section - A (खण्ड-अ)

Objective Questions (वस्तुनिष्ठप्रश्न)

5×4=20

1. Answer all the following questions.

निम्नलिखित सभी प्रश्न अनिवार्य हैं।

1. ____ means communication without words.
a. Object communication
b. Written communication
c. Oral communication
☒ d. Non-verbal communication
2. The person who transmits the message is called ____.
a. channel
☒ b. sender
c. receiver
d. response
3. ____ aims at making people work together for the common good of the organization.
☒ a. communication
b. conversation
c. combination
d. connection
4. At each stage in the process of communication, there is a possibility of interference which may hinder the process. Such interference is known as ____.
a. sender
b. receiver
☒ c. barrier
d. none of them
5. ____ describes all forms of human communication that are not verbal.
a. prosody
b. vocalics
c. haptics
☒ d. paralanguage

Course Name
Paper No.
Time -
समय -

Section - B (खण्ड-ब)
Short Answer Questions (लघु उत्तरीय प्रश्न)

2. Answer any two of the following questions in maximum 150 words.
निम्नलिखित में से किन्हीं दो प्रश्नों के उत्तर अधिकतम 150 शब्दों में दें।

2×10=20

- i) What are the common barriers that hinder effective listening?
- ii) Differentiate between a Resume, CV and Bio-Data
- iii) Write an email to send to your class teacher and HoD explaining why you will be missing online classes for a week due to health issues, that are not related to Covid.
- iv) Differentiate between definite, indefinite and distributive numerical adjectives.

Section - C (खण्ड-स)
Descriptive Questions (विवरणात्मक प्रश्न)

3. Answer any two of the following question in maximum 250 words.
निम्नलिखित में से दो किन्हीं दो प्रश्नों के उत्तर अधिकतम 250 शब्दों में दें।

2×15=30

- i. With reference to Intrapersonal Barriers to communication explain Wrong Assumption and Differing Background.
- ii. How is it important to have good presentation skills? Give at least five points.
- iii. What is a Sales Letter? As a Sale Executive draft a sales letter for promoting sales of i-pad launched by Samsung.

SEMESTER END EXAMINATION, JULY-AUGUST - 2022

Course Name:-B.Tech.

Paper Name: ENGINEERING CHEMISTRY

Time - 2 Hrs + 20 minutes per hour extra time for V.I. & examinees with writer.

Semester- II
Paper Code:- T85-2023

Max Marks-70

समय-2घंटे+20 मिनट प्रति घंटे अतिरिक्त-दृष्टिबाधित एवं सह लेखक परीक्षार्थियों के लिए। अधिकतम अंक-70

Instructions:

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- खण्ड-ब में चार प्रश्नों में से किन्हीं दो प्रश्नों के उत्तर अधिकतम डेढ़ सौ (150) शब्दों में दें। प्रत्येक प्रश्न 10 अंक का है।
- खण्ड- स में तीन प्रश्नों में से किन्हीं दो प्रश्नों के उत्तर अधिकतम ढाई सौ (250) शब्दों में दें। प्रत्येक प्रश्न 15 अंक का है।

Section - A (खण्ड-अ)

Multiple Choice Questions(बहुविकल्पीय प्रश्न)

5 x 4 = 20

1. Answer all the following questions.

निम्नलिखित सभी प्रश्न अनिवार्य हैं।

- (i) What is the relation between bond order and bond length?
- (a) Directly proportional (b) Inversely proportional
(c) Both options (a) & (b) are correct (d) None of these options are correct
- (ii) Which of the following molecule is not true about paramagnetic molecules?
- (a) Attracted by the magnetic field (b) A molecular orbital is singly occupied
(c) An example is oxygen molecule (d) Repelled by the magnetic field
- (iii) Which of the following metal salt is not responsible for permanent hardness in water?
- (a) CaSO_4 (b) MgSO_4 (c) $\text{Mg}(\text{HCO}_3)_2$ (d) CaCl_2
- (iv) Which polymer is produced by polycondensation between $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$ with $\text{COOH}-(\text{CH}_2)_4-\text{COOH}$
- (a) PVC (b) Nylon 6,6 (c) Nylon 6,10 (d) Buna-N
- (v) In UV-Visible spectroscopy Bathochromic shift is due to:

- (a) Shift of $\lambda_{(max)}$ to shorter wavelength
 (c) Shift of $\lambda_{(max)}$ to longer wavelength

- (b) Increase in intensity of $\lambda_{(max)}$
 (d) Decrease in intensity of $\lambda_{(max)}$

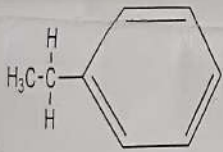
Section - B (खण्ड - ब)

Short Answer Questions (लघु उत्तरीय प्रश्न)

2. Answer any two of the following questions in maximum 150 words.

निम्नलिखित में से किन्हीं दो प्रश्नों के उत्तर अधिकतम डेढ़ सौ (150) शब्दों में दें।

2x10=20

- (i) Calculate the bond order of O_2^+ , O_2^{2-} , O_2 and arrange them in increasing order of their stability.
- (ii) Discuss the application of Infrared (IR) Spectroscopy and what is the importance of finger print region in IR- spectroscopy.
- (iii) Predict the NMR signals in the following compounds along with explanation:
- (a) 
- (b) $CH_3-CH_2-CH_2-OH$
- (c) 
- (d) $CH_3-CH_2-NH_2$
- (iv) Explain vulcanization of rubber and its advantages.

Section - C (खण्ड - स)

Long Answer Questions (दीर्घ उत्तरीय प्रश्न)

3. Answer any two of the following question in maximum 250 words.

2x15=30

निम्नलिखित में से किन्हीं दो प्रश्नों के उत्तर अधिकतम ढाई सौ (250) शब्दों में दें।

- (i) Define primary fuel and discuss in detail about Gross Calorific value and Net Calorific value. Find the Gross Calorific Value of the fuel from the following data, obtained in a bomb calorimeter experiment:

Weight of Coal burnt = 0.994 gm
 Weight of Water in calorimeter = 2592 gm
 Weight of bomb, calorimeter, stirrer = 3940 gm
 Rise in temperature of water = $2.732^\circ C$
 Mean Specific heat of the apparatus = 0.098

(ii) Describe preparation, properties and application of:

- (a) Buna-N Rubber
- (b) Nylon – 6
- (c) Polyvinyl Chloride (PVC)

(iii) Distinguish between permanent hardness and temporary hardness of water. Discuss role of Zeolite for water treatment process.